Brett Bono

Williamsburg, VA
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SUMMARY

Mathematics and software development specialist with proven expertise in algorithmic implementation and full-stack application development. Experienced in building mobile and web applications, implementing numerical algorithms, and developing data-driven solutions using React Native, Python, and Firebase

SKILLS

Python, Java, JavaScript, React, React Native, C, C++, Go (gRPC), MATLAB, Firebase, Zustand, Excel, LaTeX, Numerical Analysis, Algorithmic Implementation, Data Structures & Algorithms, Mathematical Modeling, Operations Research, SVD, PageRank, Mobile Development, Web Scraping, Real-time Data Processing

EXPERIENCE

Get Out Official — Williamsburg, Virginia

Full-Stack Mobile & Web Developer — January 2024 - May 2025

- Developed "Get Out Official," a React Native event discovery application that connects users with local events through real-time data integration and personalized recommendations.
- Built interactive map-based UI with Firebase backend and Zustand state management.
- Architected scalable microservices infrastructure supporting real-time event updates and user engagement analytics.

Williams Landscaping — Williamsburg, Virginia

Data Entry Specialist — July 2025 - Present

- Organized and digitized receipts from work orders, hardscape material suppliers, and other sources. Streamlined the file system for improved accessibility and accuracy. Supported the estimator by entering actual costs into the system to help assess estimate accuracy.

RESEARCH

James Madison University, Department of Mathematics — Harrisonburg, Virginia

Computational Mathematics Research Associate — September 2023 - December 2023

- Conducted independent research on advanced numerical algorithms and their applications in image processing and network analysis, developing optimized implementations of SVD, PageRank, and eigenvalue computation methods.
- Applied Singular Value Decomposition (SVD) techniques to cancer dataset analysis, contributing to advancements in medical analysis.
- Implemented Google PageRank algorithm and various eigenvalue methods (Power, Inverse, Rayleigh Quotient, QR Iteration) for large-scale matrix computations.

TECHNICAL PROJECTS

Automated Arbitrage Detection System

Quantitative Developer

- Engineered an automated arbitrage detection system in Java that scraped 4 major e-commerce platforms (Craigslist, eBay, Amazon, Facebook Marketplace).
- Implemented multi-threaded web scraping architecture, processing 100+ product listings to surface price discrepancies across marketplaces.

EDUCATION

James Madison University, College of Science & Mathematics — Harrisonburg, VA

Bachelor of Science in Mathematics & Minor in Computational Analytics — August 2021 - May 2025

- Presenter of "GMRES Method: Krylov Subspaces and Computational Complexity" at 2025 Mathematics Research Symposium, with accompanying academic report on algorithm performance and accuracy.
- Startup pitch competition winner with blockchain-based professional network platform, Blockchain Club (2024-2025), and member of the Competitive Programming Club (2024-2025).
- Varsity Esports athlete awarded academic scholarship in 2023 for team performance and leadership.